

Data-Driven Modeling for Robotic Mechanical Thrombectomy

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Mechanical thrombectomy has become the standard of care for acute ischemic stroke, a leading cause of death and long-term disability worldwide. However, these procedures often rely on X-ray fluoroscopic imaging, which is highly dependent on blood flow and thus fails to estimate the precise size of the clot interrupting the blood flow. This study presents a novel approach for real-time estimation of clot length based on force sensor signals and imaging data using a GRU model architecture. Promising results are shown, with models predicting the start and end of the clot with accuracies of 76.19% and 61.9%, respectively, within a 1.5 mm confidence margin and an inference time of 0.5319 ms. This method nevertheless remains a proof of concept and requires further investigation and validation.

1 Introduction

Stroke remains one of the leading causes of death and long-term disability worldwide, with a growing global burden driven by aging populations and increasing prevalence of vascular risk factors such as hypertension, diabetes, and unhealthy lifestyle behaviors. According to the World Stroke Organization Global Stroke Fact Sheet 2025, stroke accounts for a substantial proportion of global mortality and disability-adjusted life years (DALYs), with ischemic stroke representing the majority of cases and most commonly resulting from thromboembolic occlusion of cerebral arteries.¹ These occlusions interrupt cerebral blood flow, depriving brain tissue of oxygen and nutrients and triggering a cascade of ischemic injury. The burden of ischemic stroke is particularly high in low- and middle-income countries, where access to specialized stroke care remains limited, underscoring the need for effective and scalable treatment strategies.² Mechanical thrombectomy (MT) is an endovascular reperfusion therapy designed to physically remove occluding thrombi from large cerebral vessels. In contrast to intravenous thrombolysis, which delivers clot-dissolving agents systemically, MT employs devices such as stent retrievers or aspiration catheters to directly engage and extract the clot, thereby restoring cerebral blood flow. Controlled trials and subsequent meta-analyses have established MT as the standard of care for acute ischemic stroke due to large vessel occlusion when performed promptly in appropriately selected patients, demonstrating significant improvements in functional independence and reductions in long-term disability compared with medical therapy alone.² Although mechanical thrombectomy is traditionally performed

manually by interventional neuroradiologists in specialized stroke centers, recent technological advances are expanding the field toward robotic-assisted interventions. Robotic systems are being investigated for their potential to enhance procedural precision, reduce operator fatigue, and reduce procedural time. Preliminary benchtop studies using robotic platforms in modeled vasculatures suggest that robotic thrombectomy can achieve recanalization rates comparable to manual techniques, although challenges such as increased procedural time and embolic debris generation remain active areas of research.³ Despite these advances, current thrombectomy procedures still rely heavily on X-ray fluoroscopic imaging combined with the injection of radiopaque contrast agents. While this approach enables visualization of blood flow, it provides limited information beyond the occlusion site, as contrast propagation ceases after the clot due to flow interruption. As a result, critical parameters such as clot length and morphology remain difficult to assess. In practice, this uncertainty often leads clinicians to deploy stent retrievers of maximal length rather than optimally sized devices, potentially increasing the risk of vascular injury and procedural complications. These limitations highlight the need for improved real-time methods to estimate clot extent during intervention. Roussinova et al. recently proposed an approach based on force sensor data acquired during in vitro insertion experiments using a robotic mechanical guidewire.⁴ In their study, the start and end positions of the clot were predicted with accuracies of 86.5% and 71.2%, respectively, within a tolerance of ± 1.5 mm. While promising, this approach relies on curve-fitting techniques and consequently requires data from a complete insertion to infer clot length, limiting its suitability for real-time prediction. In this context, machine learning methods are attracting increasing interest due to their ability to provide rapid inference from complex intra-procedural data streams.⁵ In particular, recurrent neural networks (RNNs) are well suited for sequential data processing, as they maintain an internal memory state that captures temporal dependencies. In this report, we investigate the use of Gated Recurrent Unit (GRU) models for real-time prediction of guidewire in-clot and out-of-clot states during insertion experiments, using multimodal sensor signals and imaging data. The proposed approach aims to enable online estimation of clot boundaries and support safer, more efficient robotic thrombectomy procedures.

2 Methods

The objective of this project is to train different machine learning models on existing datasets of in vitro insertion procedures with varying experimental designs and to evaluate their ability to classify whether the guidewire is inside or outside the clot. This final time-series label of guidewire position can then be used to estimate the clot start and end positions, and thus its length. Different models are studied and described here, trained on different data modalities as well as on different datasets with distinct experimental designs.

2.1 Data Acquisition

This study leverages different existing datasets acquired by the Rheassist Lab at EPFL from in vitro thrombectomy experiments with varying experimental designs, aiming to capture variations in guidewire-clot interaction under different anatomical configurations, guidewire geometries, and insertion techniques. This section presents the data acquisition procedures for these different datasets.

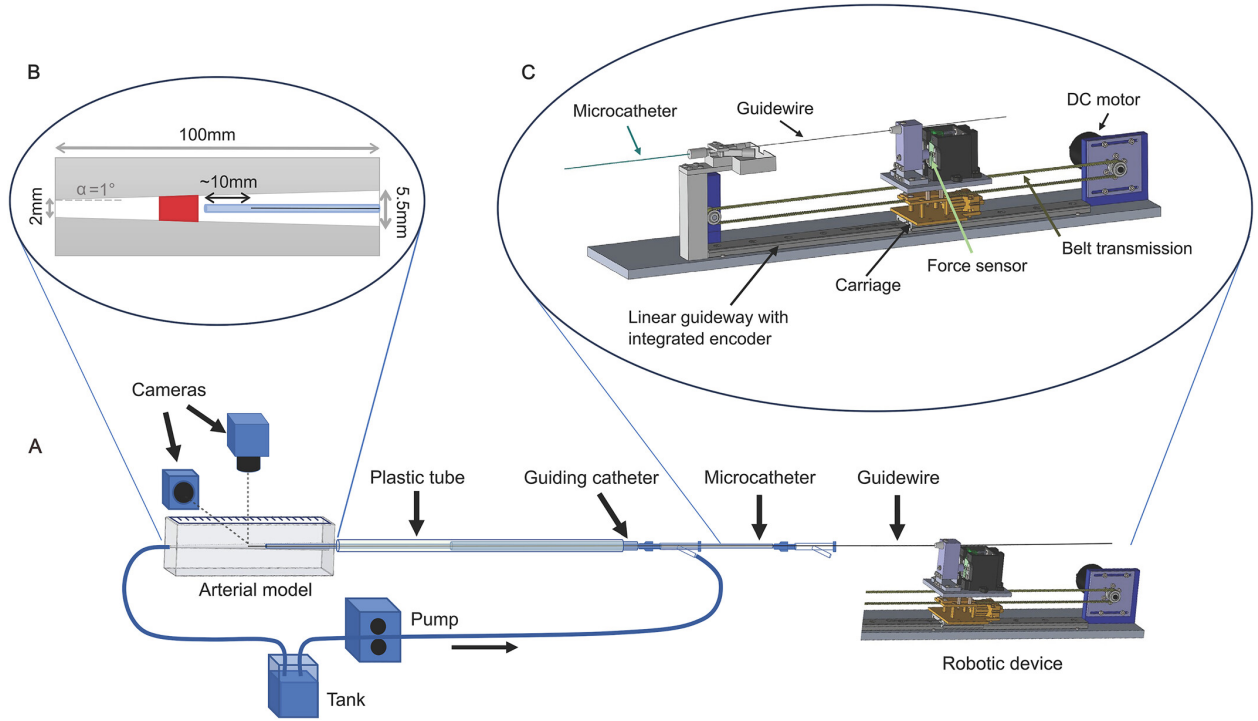


Figure 1: Diagram representing the insertion experiment with the conical arterial model and the straight guidewire.

2.1.1 Core Experimental Setup

The core experimental setup, illustrated in Figure 1, consists of a guidewire mounted on a custom robotic insertion device and advanced into a synthetic arterial model. The guidewire is inserted through a 0.53 mm inner diameter microcatheter, which is itself placed within a guiding catheter, reproducing the coaxial catheter configuration used clinically. A uniaxial force sensor mounted at the proximal end of the guidewire records axial insertion forces throughout the procedure. Prior to each insertion, the system is perfused with water using a pump to ensure consistent lubrication and fluid conditions. Multiple experimental configurations were tested by varying the arterial model geometry, guidewire shape, insertion technique, and clot presence. In addition, control insertions without a clot were performed to provide negative examples for model training and evaluation.

2.1.2 Robot

The robotic device illustrated in Figure 1C is a one-degree-of-freedom (1-DoF) translational mechanism designed to control the axial motion of the guidewire while simultaneously measuring the forces applied to it. For insertion procedures involving a twisting motion, a second robotic platform with two degrees of freedom is employed, as described later in this report. Both robotic systems are equipped with the same force sensor, the MilliNewton-B-1000-L-U-B (RYCONS Sensor Technology GmbH, Morges, Switzerland).

2.1.3 Arterial Model

Two arterial models of increasing anatomical complexity are present in the dataset.

1. The **conical** model consists of a block of stiff silicone containing a tapered cylindrical lumen (Figure 2). Silicone was selected following established practices in regulatory testing of endovascular devices (e.g., FDA bench tests). The tapered channel is 100 mm long, with a taper angle of 1° and a distal diameter of 2 mm. This geometry spans the typical diameter range of large cerebral vessels commonly involved in acute ischemic stroke, including the internal carotid artery (ICA) and the M1–M2 segments of the middle cerebral artery (MCA).
2. The **anatomical** model represents the MCA region, including both M1 and M2 branches (Figure 3). This model introduces curvature and bifurcations, allowing assessment of whether models trained on simplified geometries generalize to anatomically realistic conditions.

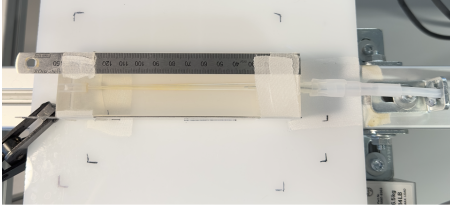


Figure 2: Top view of the conical arterial model.

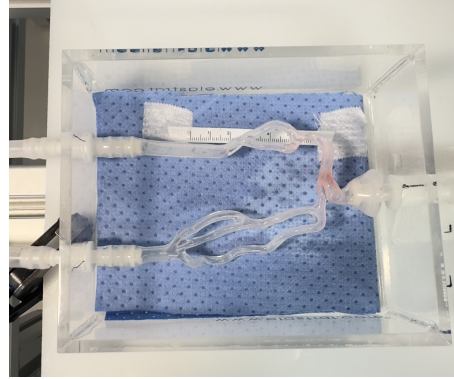


Figure 3: Top view of the anatomical arterial model.

2.1.4 Clot

Guar gum and borax hydrogel cylinders have been produced and used as synthetic clot analogs to produce the experimental dataset. Three stiffness categories were produced, soft, medium, and hard, by varying the hydrogel composition. In the conical model, the clot was consistently placed at a fixed axial position to reduce variability across experiments. In the anatomical model, two clot locations were considered: a proximal placement in the M1 segment and a distal placement near the M2 branches. These configurations were chosen to evaluate the effect of vessel geometry and access difficulty on model performance. Unless otherwise stated, clot stiffness and placement were not explicitly encoded as model inputs; instead, the models were trained to infer clot state directly from sensor and image data.

2.1.5 Guidewire

The Avigo™ (Medtronic, Irvine, CA, USA) guidewire (0.36 mm diameter) was used in two configurations:

1. **Straight**, with no changes from the industrial wire.
2. **Bent**, manually shaped to 45° by pinching it slightly and sliding along it using an insertion tool.

The bent guidewire more closely reflects clinical practice, where shaped tips are commonly used to facilitate navigation and clot penetration, and was therefore used in the most realistic experimental configurations.



Figure 4: Bent guidewire with a 45° angle.

2.1.6 Insertion Technique

Two different insertion techniques were used:

1. **No Twist**: This technique consists of a pure translational motion at a fixed velocity of 1 mm/s.
2. **Twist**: This insertion technique controls both translational and rotational velocities, which are adjusted proportionally to the force measured by the sensor. When the measured force exceeds $T_1 = 10$ mN, the guidewire starts rotating. At a force threshold of $T_2 = 20$ mN, the translational velocity is progressively reduced. When the force reaches $T_3 = 30$ mN, the guidewire rotates at its maximum rate while the translational velocity is reduced to zero. The maximum translational velocity is set to $v_{\text{transl,max}} = 1$ mm/s, and the maximum rotational velocity is $v_{\text{rota,max}} = 180^\circ/\text{s}$.

In all insertion procedures, both **forward** and **backward** directions were recorded. However, for some insertions, one of the two directions was discarded due to malfunctions, most often caused by buckling of the guidewire during penetration attempts.

2.1.7 Datasets

Each insertion experiment is characterized by a combination of arterial model, clot placement, guidewire shape, insertion technique, and clot presence. Two datasets are considered in this study: a **base dataset**, previously used by Roussinova et al., which can be seen in Table 1 and an additional dataset including both data excluded from the Roussinova et al. (2025) paper and data acquired by Yucef Grebici during his Master’s thesis.^{4,6} The latter dataset is further referred as **additional dataset** and detailed in Table 2. A single count represents either a forward or backward insertion experiment.

Table 1: Base Dataset Configuration Counts

Model	Placement	Wire Shape	Technique	Clot Presence	Count
Conical	Conical Placement	Straight	No Twist	With	208

Table 2: Additional Data Characteristics (Counts With / Without Clot)

Model (Total)	Placement (Total)	Wire Shape (Total)	Technique	Count (W. / Wo.)
Anatomical (181)	Bifurcation (73)	Bent (28)	No Twist	22 / 6
		Straight (45)	No Twist	39 / 6
	Proximal (108)	Bent (72)	No Twist	20 / 6
		Straight (36)	Twist	46 / 0
Conical (233)	Conical Placement (233)	Bent (193)	No Twist	30 / 6
			Twist	76 / 16
		Straight (40)	Twist	101 / 0
			No Twist	34 / 6

2.2 Data Description

During the experiments described above, multimodal data were collected, including force sensor measurements, synchronized image recordings, and annotations marking the start and end of guidewire–clot contact retrieved from the post-processing of images. This section describes the characteristics of each data modality, further in the report, the data preprocessing section outlines the preprocessing steps applied to the different data modalities prior to model training and evaluation.

2.2.1 Sensor Data

Axial insertion force was recorded at 1 kHz using a MilliNewton-B-1000-L-U-B force sensor (RYCONS Sensor Technology GmbH, Morges, Switzerland). The high sampling rate allows capture of transient force variations associated with guidewire–clot interaction. An example of a raw force signal is shown in Figure 5.

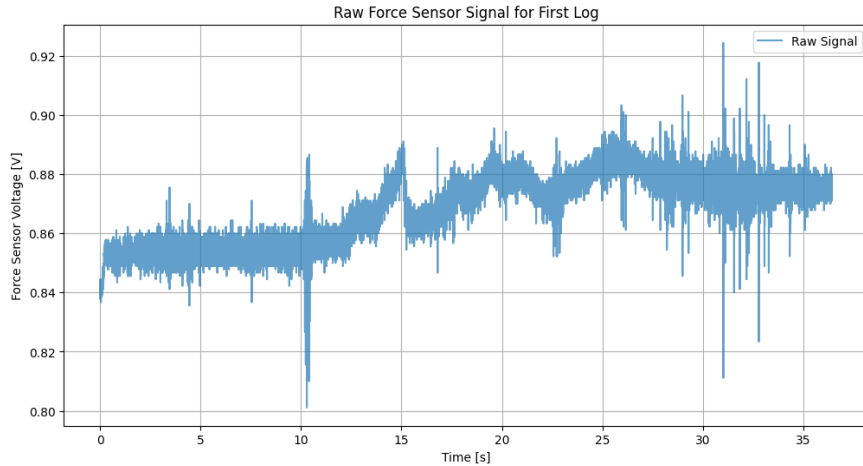
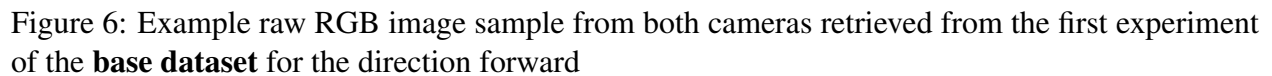
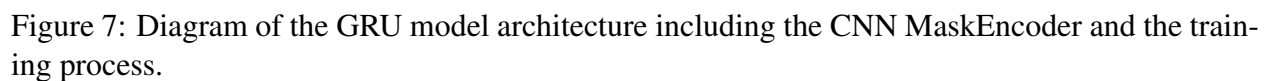


Figure 5: Example raw signal retrieved from the first experiment of the **base dataset** for the direction forward

Two RGB cameras recorded the insertion process at 10 Hz from top and side viewpoints, providing complementary visual information about guidewire deformation and lateral displacement. The model VCXU-124C (Baumer GmbH, Friedberg, Germany) was used, with varying capture regions of the pixel matrix depending on the requirements, resulting in different resolutions and aspect ratios. An example image pair is shown in Figure 6.



The proposed prediction model can take as input either one or both signal modalities, namely the raw signal with optional additional bandpower channels and a sequence of binary image data. It outputs a sequence of the same length as the input, corresponding to the label sequence described above, which indicates whether the guidewire is inside or outside the clot over time.



The model architecture works for any sequence length thanks to its recursive aspect, applying to each element of the sequence the same weights while keeping an updated memory state. Addi-

tionally the GRU implementation of RNN allows to reduce the effects of vanishing or exploding gradient which is often a common issue in RNN training in addition to being faster at training time than LSTM. The GRU Unit takes as parameters `hidden_size` representing the size of the memory hidden state and `num_layers` representing the number of GRU Unit applied sequentially. At the initial time step $t = 0$, the hidden state is initialized to zero: $h_0 = \mathbf{0}$.

For each time step $t \geq 1$, the Gated Recurrent Unit (GRU) updates its hidden state according to the following equations:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z), \quad (1)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r), \quad (2)$$

$$\hat{h}_t = \phi(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h), \quad (3)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t. \quad (4)$$

In these equations, $\sigma(\cdot)$ denotes the sigmoid activation function, $\phi(\cdot)$ typically represents the hyperbolic tangent activation, and $\odot$ denotes element-wise multiplication.

Let d denote the number of input features and e the size of the hidden state. The variables and parameters are defined as follows:

- $x_t \in R^d$: input vector at time step t
- $h_t \in R^e$: hidden state (output) vector
- $\hat{h}_t \in R^e$: candidate activation vector
- $z_t \in (0, 1)^e$: update gate vector
- $r_t \in (0, 1)^e$: reset gate vector
- $W_* \in R^{e \times d}$: input weight matrices
- $U_* \in R^{e \times e}$: recurrent weight matrices
- $b_* \in R^e$: bias vectors

All parameters W_* , U_* , and b_* are learned during training.

In order to produce one binary value from this hidden state representing in-clot or not a final linear layer is applied from `hidden_size` to 1 which is then passed through a sigmoid function return values between 0 and 1 representing probabilities of being in clot

2.3.2 MaskEncoder

The `MaskEncoder` takes as input an image tensor with `in_channels` fixed to 2, corresponding to the two distinct camera views, and produces a compact embedding of dimension `emb_dim`. To limit model complexity, reduce the risk of overfitting, and significantly decrease training time, the encoder relies on a deliberately simple convolutional neural network architecture. The architecture consists of a single convolutional layer with 8 output channels and a kernel size of 3×7, followed by batch normalization and a ReLU nonlinearity. This convolution captures local spatial features

while accounting for the elongated geometry of the guidewire motion. Spatial dimensions are then collapsed using global average pooling, resulting in a fixed-size feature vector independent of the input resolution. Finally, a fully connected layer projects this vector into a low-dimensional embedding space of size `emb_dim`. The resulting embedding is used as a compact representation of the image input and can be directly integrated into the subsequent temporal model.

2.4 Data Preprocessing

As discussed, the GRU architecture requires sequential data with the shape (sequence length, number of channels). Each channel therefore needs to be temporally aligned, representing information occurring at the same time, whether it is a force signal, frequency information, or a captured image. Additionally, images are two-dimensional, and flattening them would result in a loss of spatial information. Consequently, a MaskEncoder based on a CNN architecture is used to encode the 2D images into a 1D representation. The following steps describe how this preprocessing was performed.

2.4.1 Temporal Features

Raw force signals were low-pass filtered at 5 Hz to remove high-frequency noise unrelated to macroscopic guidewire motion. The filtered signal was then downsampled to 10 Hz to match the camera acquisition rate. Each sequence was standardized on a per-experiment basis. To prevent the model from learning a trivial temporal bias, since clot contact typically occurs around 10 s, each sequence was randomly trimmed by removing 1–5 s from the beginning, while ensuring at least 2 s of pre-insertion data.

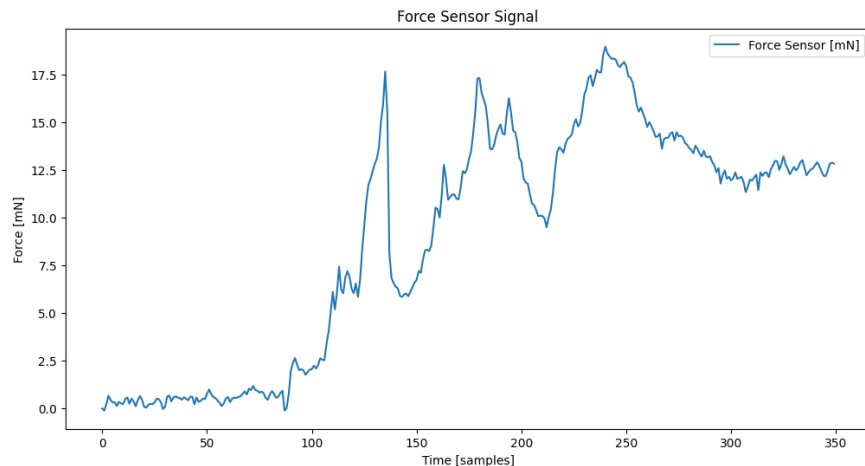


Figure 8: Example of preprocessed force signal from a base dataset insertion.

2.4.2 Frequency-Domain Features

In addition to the raw force signal, frequency-domain features were extracted to capture higher-frequency force fluctuations. The raw signal was notch-filtered at 50 Hz and its harmonics to

remove electrical noise. Ten bandpower features, evenly distributed between 0 and 500 Hz, were computed using sliding windows of 500 samples with a step size of 100 samples, resulting in an effective feature rate of 10 Hz and a prediction delay of 0.5 s.

2.4.3 Image Preprocessing and Mask Generation

The acquired RGB images were first converted to grayscale and rotationally aligned according to each camera position to ensure spatial consistency across views, such that the guidewire consistently moves from left to right during insertion. The images were subsequently downsampled by a factor of four to reduce signal dimensionality and computational complexity.

At each time step t , the image input is intended to represent visual information corresponding to changes occurring at that time step, rather than the full scene state. After background subtraction, the image primarily captures incremental guidewire motion, such as local lateral displacement or torsional deformation induced by interaction forces, which may differ depending on whether the guidewire is inside or outside the clot.

To extract these dynamic features while simplifying the image representation and limiting computational cost—given the heavy training requirements of the GRU—an exponential moving average was applied:

$$\mathbf{I}_t = \alpha \mathbf{X}_t + (1 - \alpha) \mathbf{I}_{t-1}, \quad (5)$$

where $\mathbf{X}_t$ denotes the preprocessed image at time t , $\mathbf{I}_t$ the exponentially averaged background image, and $\alpha \in (0, 1)$ the smoothing factor. This procedure maintains an adaptive background model that is subtracted from the current image to isolate changes occurring at time t . The resulting difference image emphasizes instantaneous guidewire motion while suppressing static background content, thereby simplifying the learning task for the GRU by providing a compact, motion-based representation rather than an absolute configuration of the scene. A smoothing factor of $\alpha = 0.01$ was used, and the difference image was lower-thresholded at 0.05 to obtain a binary mask representing regions of motion at the current time step. The resulting masks are highly sparse, with large regions of zero-valued pixels corresponding to areas outside the arterial model. To remove these irrelevant regions, cropping bounds were computed as the spatial region in which at least one pixel was active in any sample of a given experiment, with an added margin of 20 pixels. The most extreme bounds across all insertions were then used to define a common cropping region, ensuring that all relevant motion information was retained. After cropping, the images had approximate dimensions of 60×300 , with slight variations depending on the camera used. To obtain a consistent two-channel image suitable for input to a convolutional encoder, all images were interpolated to a fixed resolution of 60×300 . Higher resolution was preserved along the lateral direction, reflecting the dominant direction of guidewire insertion. The camera acquisition was not hardware-synchronized with the force sensor. Consequently, simply downsampling the force signal to the image frame rate does not guarantee temporal correspondence between modalities, and some image frames were missing due to intermittent camera malfunctions. To address this, the camera trigger signal recorded at 1000 Hz was used to reconstruct image timestamps. In insertions where the camera was triggered later than the force sensor, the force signal was cropped to begin at the first camera timestamp. A final synchronized multimodal sequence was constructed by associating each force sensor timestamp with the temporally closest available image frame.

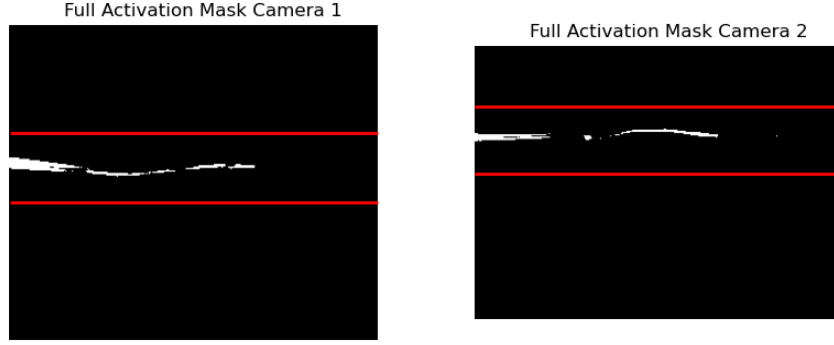


Figure 9: Cumulative mask for both camera 1 and camera 2 constructed by setting mask to true if true in any mask frame of the insertion process. This allows to visualize the complete information of an insertion obtained after preprocessing the images.

2.4.4 Label preprocessing

For each experiment both start and end times were recorded as the time where the guidewire enters in contact with the clot and exits the clot respectively. A binary label sequence was built for each experiment matching the sequence of data previously retrieved with 1's between start and end timestamps and 0 otherwise. For Experiment without clot the label is simply a sequence of zeros of the same length as the sensor sequence. An example of this label can be seen in Figure10. While most clots have a length of approximately 10 mm, fewer samples exhibit significantly longer clots. This imbalance is taken into account during the training process and is discussed in the corresponding section.

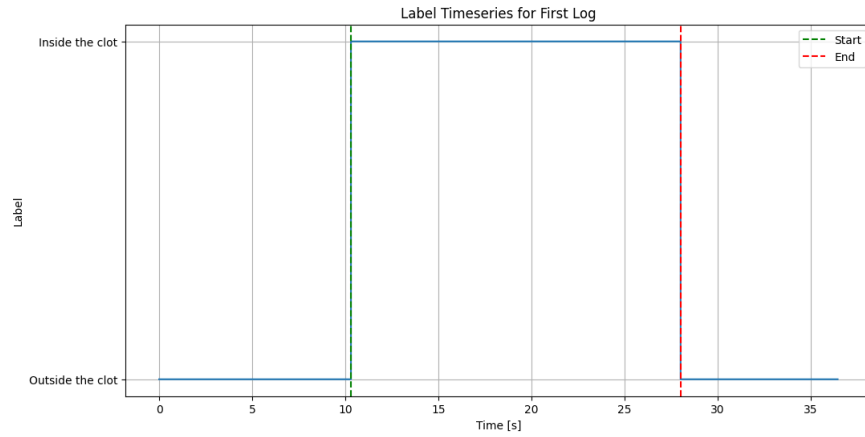


Figure 10: Diagram of the true label which the designed RNN model will be trained to predict. This label represent the presence of the guidewire in or outside of the clot at each time during the insertion procedure.

2.5 Training

Training was done using Adam optimizer and Binary Cross Entropy loss (BCEloss) defined as :

$$\ell(\mathbf{x}, \mathbf{y}) = \mathbf{L} = [l_1 \ \dots \ l_N]^\top, \quad (6)$$

$$l_n = -w_n [y_n \log(x_n) + (1 - y_n) \log(1 - x_n)], \quad n = 1, \dots, N. \quad (7)$$

with

- $x_n \in (0, 1)$: predicted probability
- $y_n \in \{0, 1\}$: ground-truth label
- $w_n \in \mathbb{R}^+$: sample weight

Additionally, sample weights were introduced to compensate for the imbalance in clot length distribution. To balance the learning process with respect to clot length, a backpropagated weight was computed for each experiment.

Let ℓ_n denote the clot length of sample n . The empirical distribution of clot lengths was modeled using a Gaussian Kernel Density Estimation (KDE), yielding a probability density function $p(\ell)$. The sample weight w_n was defined as the inverse of this estimated density:

$$w_n = \frac{1}{p(\ell_n)}. \quad (8)$$

To ensure numerical stability and preserve a consistent loss scale, the weights were normalized by their mean value:

$$\tilde{w}_n = \frac{w_n}{\frac{1}{N} \sum_{i=1}^N w_i}. \quad (9)$$

Finally, the normalized weights were squared to amplify the contribution of rare clot lengths during training: $\hat{w}_n = \tilde{w}_n^2$. These weights were then applied during backpropagation through the weighted binary cross-entropy loss. The resulting weights of the training subset of the **base dataset** can be seen in Figure 11.

Finally, weight decay and early stopping were introduced to reduce overfitting. Early stopping used 20% of the training set for validation; this same split was also used in the cross-validation process described later. The early stopping criterion was controlled by a patience parameter, which dictated the number of epochs allowed without improvement in the F1 score. All models were trained for a maximum of 500 epochs.

2.6 Evaluation

The evaluation of each model was conducted on multiple subsets of the available data and consisted of three main metrics: (i) the overall mean loss, (ii) the F1 score computed between the true and predicted labels after lower-thresholding the predictions at 0.9 (all values ≥ 0.9 are set to 1, and all others to 0), and (iii) the accuracy of the predicted start and end times. The start and end times are defined as the first transitions from 0 to 1 and from 1 to 0, respectively. Start and end accuracy were

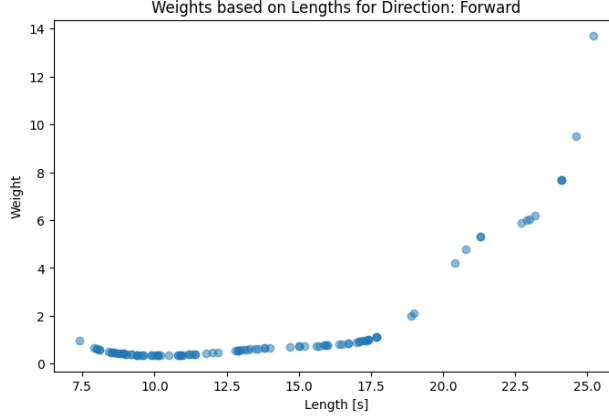


Figure 11: Weights and lengths of the training portion of the base dataset.

computed using the same procedure as proposed by Roussinova et al., with a tolerance margin of ± 1.5 s. For experiments without a clot, the start and end times are defined as *None*; such cases are therefore considered correctly classified only if no 0-to-1 or 1-to-0 transitions occur in the predicted labels. Additionally, a baseline prediction was used for comparison. It was constructed by computing the mean start and end indices from the training dataset. The baseline model then predicts a sequence of the appropriate length, assigning a value of 1 between the mean start and end indices and 0 elsewhere.

2.7 Cross Validation

For all experiments described below, a grid search combined with k-fold cross-validation was performed to identify the optimal hyperparameters. A validation subset corresponding to 20% of the training data was used during training. The model achieving the best average validation performance across folds was retained and subsequently used for evaluation on the test set. When no image data were included in the input features, the hyperparameter search space shown in Table 3 was used, with k=5 folds for cross-validation. When image data were present, the search space was restricted in order to limit computational cost and reduce overfitting. In particular, the embedding dimension of the image encoder was introduced as an additional hyperparameter, while the recurrent model depth and learning parameters were constrained. The corresponding search space is reported in Table 3.

Table 3: Hyperparameter search space for models trained with and without image input.

Hyperparameter	Without Images	With Images
Learning rate	$\{10^{-4}, 10^{-3}\}$	10^{-3}
Weight decay	$\{10^{-5}, 10^{-4}\}$	10^{-4}
Hidden size	$\{8, 16, 32\}$	$\{8, 16\}$
Number of layers	$\{1, 2\}$	1
Embedding dimension (emb_dim)	—	$\{1, 2\}$
Patience	50	100

3 Model Evaluation Across Data Subsets and Modalities

This section presents the systematic evaluation of the proposed model architecture under different data configurations. Across all evaluations, the same model architecture and training procedure were used; only the training and test datasets, as well as input modalities, varied. Tables are used to summarize data splits and configurations for clarity and reproducibility. The different input modalities, resulting in 7 distinct feature sets, are listed in Table 4.

Table 4: Data configurations used in the base dataset evaluation

Config ID	Features	Direction
C1	Signal	Forward
C2	Signal + Bandpower	Forward
C3	Images	Forward
C4	Signal + Bandpower + Images	Forward
C5	Signal + Bandpower + Images	Backward
C6	Signal + Bandpower + Images	Both
C7	Signal	Both

3.1 Base Dataset Evaluation

The first evaluation was performed on the base dataset, to provide a comparison to results published by Roussinova et al. Unlike their curve-fitting method, which allows testing on all samples, our machine learning approach requires a train/test split. Accordingly, 80% of the dataset was used for training and validation, and the remaining 20% for testing. Training and validation splits are reported in Table 5, and test sets are listed in Table 6.

Table 5: Training and validation datasets for base dataset evaluation. Forward/backward sample counts are reported as F/B.

Model	Guidewire	Technique	Clot	Train (F/B)	Val (F/B)
Conical	Straight	No Twist	With	66 / 66	17 / 17

Table 6: Test datasets for base dataset evaluation. Forward/backward sample counts are reported as F/B.

Name	Model	Guidewire	Technique	Clot	Test (F/B)
Base Dataset (test)	Conical	Straight	No Twist	With	21 / 21
Conical Bent	Conical	Bent	No Twist	With	38 / 38
Anatomical	Anatomical	Straight	No Twist	With	35 / 34
Conical No Clot	Conical	Straight	No Twist	Without	3 / 3

3.2 Bent Guidewire Evaluation

To assess the impact of a 45-degree bent guidewire, models were trained and validated on bent-wire insertions. Training and validation splits are reported in Table 7, and test datasets are listed in Table 8.

Table 7: Training and validation datasets for bent guidewire evaluation. Forward/backward sample counts are reported as F/B.

Model	Guidewire	Technique	Clot	Train (F/B)	Val (F/B)
Conical	Bent	No Twist	With	23 / 24	6 / 7

Table 8: Test datasets for bent guidewire evaluation.

Name	Model	Guidewire	Technique	Clot	Test (F/B)
Conical Bent (test)	Conical	Bent	No Twist	With	9 / 7
Base Dataset	Conical	Straight	No Twist	With	104 / 104
Anatomical	Anatomical	Bent	Twist	With	35 / 34
Conical Bent No Clot	Conical	Straight	No Twist	Without	8 / 8

3.3 Twist Technique Evaluation

Models trained exclusively on twist insertions were evaluated to assess generalization to realistic thrombectomy procedures. Training and validation splits are summarized in Table 9, and test datasets in Table 10.

Table 9: Training and validation datasets for twist technique evaluation. Forward/backward sample counts are reported as F/B.

Model	Guidewire	Technique	Clot	Train (F/B)	Val (F/B)
Conical	Bent	Twist	With	35 / 28	9 / 8

3.4 Evaluation on Complete Dataset

Finally, simpler model configurations using only feature sets C1 and C7 were trained on the complete dataset and evaluated on both a held-out portion of that dataset and on the base dataset. This assessment measures whether the model can generalize across multiple insertion techniques and, in the case of C7, multiple insertion directions. Training used 132 forward insertions, while evaluation was performed on 33 forward insertions from the additional dataset and 104 from the base dataset (totaling 137 test samples).

Table 10: Test datasets for twist technique evaluation. Forward/backward sample counts are reported as F/B.

Name	Model	Guidewire	Technique	Clot	Test (F/B)
Conical Twist	Conical	Bent	Twist	With	8 / 13
Conical Straight	Conical	Straight	No Twist	With	17 / 17
Anatomical Twist	Anatomical	Bent	Twist	With	23 / 23
Conical No Clot	Conical	Straight	No Twist	Without	3 / 3
Base Dataset	Conical	Straight	No Twist	With	104 / 104

4 Results

4.1 Base Dataset Models

The results obtained with the force-only configuration (C1) show performance on the base dataset that is close to the values reported by Roussinova et al., achieving start and end accuracies of 76.19% and 61.9%, respectively, compared to 86.5% and 71.2% in their study. Across all evaluated models, predicting the end of the clot—corresponding to the guidewire exiting the clot—proved consistently more challenging than predicting clot entry, with no configuration achieving higher accuracy on end detection than on start detection, regardless of the data modality or whether forward or backward insertion sequences were used. The highest start detection performance was achieved by configuration C2, which outperformed the results of Roussinova et al. with a start accuracy of 95.23%. This improvement, however, came at the cost of reduced end detection performance. Although the proposed model architecture is able to learn from preprocessed image data (see Table 11, configuration C3), the inclusion of visual information does not lead to a significant improvement in prediction performance and in several cases degrades it (see Table 11, configuration C4). Configuration C6 demonstrates a limited ability to generalize across forward and backward insertion sequences, with performances greater than the baseline but lower than the C1 configuration. Finally, evaluation across different experimental setups indicates that the proposed architecture can generalize to alternative insertion conditions, often achieving performance exceeding the baseline. However, none of the evaluated models perform reliably when predicting insertions without a clot.

Table 11: Performance of different model configurations on the test datasets. Each row corresponds to a model defined in Table 4. Values above baseline are shown in green.

Model	Test Dataset											
	Base Dataset (test)			Conical Bent			Anatomical			Conical No Clot		
	F1	Start	End	F1	Start	End	F1	Start	End	F1	Start	End
C1	0.9244	0.7619	0.6190	0.8951	0.8947	0.2894	0.5711	0.9428	0.1142	0.0	0.0	0.333
C2	0.8669	0.9523	0.4761	0.8079	0.5526	0.1842	0.6593	0.9142	0.3428	0.0	0.0	0.33
C3	0.8059	0.7619	0.1428	0.8158	0.5	0.3684	0.5489	0.4571	0.1142	0.0	0.0	0.0
C4	0.8784	0.7619	0.4285	0.8014	0.6052	0.1578	0.5854	0.6857	0.1428	0.0	0.0	0.33
C5	0.8110	0.5714	0.4285	0.6819	0.4736	0.2368	0.3879	0.1470	0.2647	0.0	0.0	0.0
C6	0.8813	0.66	0.5952	0.7917	0.4605	0.3026	0.5223	0.5217	0.2753	0.0	0.0	0.33
Baseline	0.7411	0.2619	0.2381	0.7970	0.5132	0.3026	0.5031	0.1449	0.1159	0.0	0.0	0.0

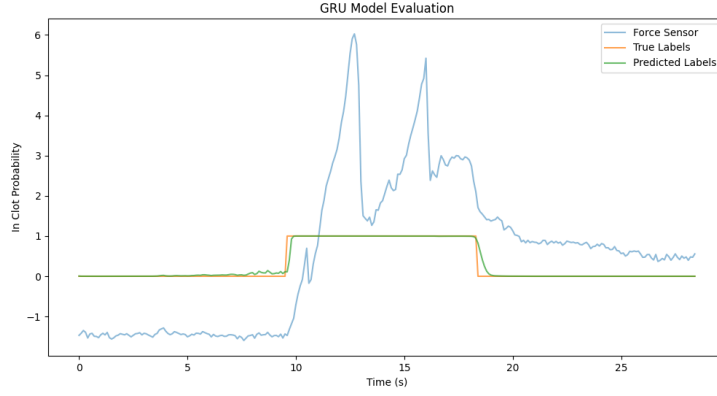


Figure 12: Example of a prediction from the C1 model trained on the Base dataset on one of the sample of the test portion of the base dataset. In blue, the standardized force sensor, input data of the model, in green the predicted label and in orange the true label.

4.2 Bent Guidewire

The results in Table 12 show that the proposed models can also learn from more complex insertion experiments, sometimes achieving even higher accuracy than in the base dataset models (100% start and 77% end accuracies for the C1 model). Observations from the base dataset models, however, also apply here, with no improvements resulting from the inclusion of image data and continued difficulty in predicting clot exit as well as insertion procedures without clots.

Table 12: Performance of different model configurations on the test datasets. Each row corresponds to a model defined in Table 4. Values above baseline are shown in green.

Model	Test Dataset											
	Conical Bent (test)			Base Dataset			Anatomical			Conical Bent No Clot		
	F1	Start	End	F1	Start	End	F1	Start	End	F1	Start	End
C1	0.9429	1.0	0.77	0.7967	0.8173	0.2980	0.7411	0.8571	0.3142	0.0	0.125	0.375
C2	0.9227	1.0	0.77	0.6604	0.5673	0.2884	0.4730	0.6	0.0857	0.0	0.0	0.0
C3	0.9131	1.0	0.66	0.5598	0.1730	0.2307	0.4602	0.0285	0.0571	0.0	0.5	0.5
C4	0.9106	0.77	0.66	0.6776	0.6442	0.25	0.7114	0.7714	0.1428	0.0	0.0	0.0
C5	0.7219	0.4285	0.2857	0.4602	0.1538	0.0288	0.5340	0.0588	0.3529	0.0	0.0	0.0
C6	0.8334	0.625	0.5	0.5982	0.3605	0.2067	0.5944	0.4057	0.1159	0.0	0.0	0.125
Baseline	0.8143	0.3750	0.4375	0.6869	0.2788	0.1250	0.5506	0.1594	0.2899	0.0	0.0	0.0

4.3 Twist Technique

The results of the twist technique models evaluation, presented in Table 13, reveal substantially weaker generalization across model configurations, with only the force-only C1 model consistently outperforming the baseline across all datasets. In contrast, the C1 configuration exhibits strong generalizability, achieving even higher performance on the conical straight test set than on the twist test set itself. In contrast, the image-only C3 configuration yields very low prediction accuracy, performing well below the baseline. Consistent with observations from the base dataset models, the inclusion of image data does not lead to performance improvements in this setting. Furthermore, models continue to struggle with accurately predicting clot exit and with correctly identifying insertion procedures without clots.

Table 13: Performance of different model configurations on the test datasets. Each row corresponds to a model defined in Table 4. Values above baseline are shown in green.

Model	Test Dataset														
	Conical Twist			Conical Straight			Anatomical Twist			Conical No Clot			Base Dataset		
	F1	Start	End	F1	Start	End	F1	Start	End	F1	Start	End	F1	Start	End
C1	0.7457	0.875	0.25	0.8458	1.0	0.6470	0.7200	0.7391	0.1739	0.0	1.0	1.0	0.6770	0.7980	0.2692
C2	0.7470	0.625	0.0	0.5419	0.0588	0.0	0.6688	0.5217	0.0869	0.0	0.0	0.666	0.5797	0.0	0.02884
C3	0.0	0.0	0.0	0.0086	0.0	0.0588	0.0	0.0	0.0	0.0	1.0	1.0	0.0004	0.0	0.0
C4	0.6786	1.0	0.25	0.5661	0.1176	0.0	0.5758	0.3478	0.0434	0.0	0.0	0.0	0.5899	0.1923	0.0288
C5	0.6368	0.3846	0.5384	0.0	0.0	0.0	0.2436	0.0869	0.0	0.0	1.0	1.0	0.0	0.0	0.0
C6	0.5986	0.4285	0.2857	0.4627	0.029	0.1176	0.070	0.0869	0.1521	0.0	0.5	0.833	0.4116	0.0576	0.0384
Baseline	0.5164	0.2381	0.1905	0.7998	0.5000	0.4118	0.4808	0.3478	0.1522	0.0	0.0	0.0	0.6534	0.3942	0.1587

4.4 Complete Data & Generalization

The final results presented in Table 14 demonstrate possible generalization across all types of insertion procedures discussed in this report, although they achieve lower accuracy than the C1 model trained on the base dataset (71.15% start accuracy and 43.26% end accuracy). Additionally, training on both forward and backward data shows an increase in end accuracy at the expense of lower start accuracy.

Table 14: Performance of different model configurations on the test datasets. Each row corresponds to a model defined in Table 4. Values above baseline are shown in green.

Model	Test Dataset					
	Additional Dataset (Test Set)			Base Dataset		
	F1	Start	End	F1	Start	End
C1	0.8254	0.8409	0.2727	0.8096	0.7115	0.4326
C7	0.7269	0.5903	0.4337	0.5808	0.3125	0.2980
Baseline	0.5831	0.1928	0.2289	0.6017	0.1346	0.1731

4.5 Inference Time

To assess the feasibility of deploying the proposed models for real-time prediction, the inference time was measured. Since inference time scales with sequence length, the average prediction time over complete insertion sequences from the additional dataset was used as an upper-bound estimate. Using this protocol, the simpler C1 model predicts an entire insertion process with a mean inference time of 0.5319 ms. This means that the model could easily be used at a 10Hz real-time feedback.

4.5.1 Computational Setup

Inference time measurements were performed on the Izar high-performance computing cluster at EPFL. All measurements were obtained on a single compute node equipped with two Intel Xeon Gold CPUs (2.1 GHz, 20 cores per processor) and an NVIDIA V100 GPU with 32 GB of memory. Each inference job was allocated one GPU, four CPU cores, and 32 GB of RAM. Inference was performed sequentially on complete insertion sequences with a batch size of one, reflecting the intended real-time deployment scenario.

5 Discussion

This work studied the feasibility of real-time clot state prediction during robotic mechanical thrombectomy using data-driven recurrent neural networks trained on multimodal *in vitro* data. Across a wide range of experimental configurations, including variations in arterial geometry, guidewire shape, insertion technique, and insertion direction, the results consistently demonstrate that simpler models trained exclusively on force-based signals outperform more complex multimodal approaches for predicting guidewire–clot interaction. With an inference time of 0.5319 ms and an f1 score of 0.92 at predicting guidewire–clot contact, the proposed `GRUClassifier` represents a promising solution for real-time estimation of clot length and mechanical properties, with potential to improve robotic mechanical thrombectomy procedures. The lack of performance improvement resulting from the inclusion of additional data modalities observed in this study does not imply that these modalities lack relevant information about guidewire–clot interaction. Indeed, image-only models still achieve performance above the baseline. This observation is most likely attributable to the limited hyperparameter search space and the use of a grid-search strategy. Training models that incorporate image data substantially increases computational cost, with some training runs lasting up to two hours per cross-validation fold; consequently, a restricted hyperparameter search space was adopted.

In addition, the performance of all models on the `No Clot` data suggests a form of overfitting to the typical insertion procedure. In these experiments, despite random temporal cropping, the guidewire generally comes into contact with the clot at approximately 8,s (around 80 samples), remains within the clot for roughly 10,s (about 100 samples), and then exits without further clot contact. This observation highlights an important limitation: the strong dependence of the proposed technique on the data distribution used for training.

Although the complete models (C1 and C7) demonstrate promising performance across substantially different experimental configurations, their accuracy remains relatively low, indicating that a significantly larger and more diverse dataset—along with more randomized experimental

protocols—would be required to achieve reliable predictions. Interestingly, the simpler C1 models trained on all different subsets appear to have learned latent information within the force signal that is transferable across all proposed experimental conditions (e.g., bent guidewire, twist insertion techniques), consistently achieving scores above the baseline. This suggests that the C1 model does not overfit to a specific training configuration and captures more generalizable force signatures of guidewire interaction. This effect is also evident in the twist models results, where the model trained on conical bent insertions with a twist technique achieves better performance on conical straight data than on the modality it was trained on. Insertions performed with a straight guidewire are inherently easier to predict, as reflected by the higher baseline accuracies, and consequently the model attains superior performance on these simpler experimental conditions.

Overall, these findings support the viability of recurrent force-based models for online clot boundary estimation while also highlighting key challenges related to multimodal integration, clot exit detection, and generalization to complex insertion behaviors.

6 Limitations and Future Steps

The proposed solution, although reaching high accuracies of sometimes 100%, shows, through its very high F1-score baseline, the limits of the experimental design of in-vitro thrombectomy. In fact, machine learning models are highly dependent on the data on which they are trained and would require much more complex and generalized data to be reliable in clinical settings. Future steps could involve the inclusion of a significant number of `No Clot` insertions in the training data, as well as optimization of the training process by, for example, batching. In fact, batching requires samples of the same length, which is not always the case with insertions of different durations. Windowing could allow both speeding up the process and reducing the model’s tendency to learn the basic insertion pattern (out of the clot, in, out). This improvement, alongside Bayesian optimization instead of grid search, could allow for more extensive hyperparameter optimization, which could include exponential moving average (EMA) parameters and thus potentially improve the results of image-based models. Finally, alternative approaches to CNNs could be used for encoding the masks, such as using the centroid positions of the N largest clusters in the image, which would ease the training process of the GRU.

7 Acknowledgments

I would like to express my sincere gratitude to William Galand and Evgenia Roussinova for their continuous guidance, insightful discussions, and constant support throughout this project.

8 Code

The code for this project can be found on github at : <https://github.com/oskarboesch/TREX-Semester-Project.git>

9 Additional Figures and tables

9.1 Hyperparameter Tables

Table 15: Model configurations, and optimal hyperparameters.

Experiment	Config	emb_dim	hidden	layers	lr	wd	patience
Twist	C1	—	32	1	10^{-3}	10^{-4}	—
	C2	—	16	2	10^{-4}	10^{-4}	—
	C3	1	8	1	10^{-3}	10^{-4}	100
	C4	1	8	1	10^{-3}	10^{-4}	100
	C5	1	8	1	10^{-3}	10^{-4}	100
	C6	2	8	1	10^{-3}	10^{-4}	100
Base	C1	—	16	2	10^{-3}	10^{-5}	—
	C2	—	16	2	10^{-4}	10^{-4}	—
	C3	2	16	1	10^{-3}	10^{-4}	100
	C4	2	16	1	10^{-3}	10^{-4}	100
	C5	1	8	1	10^{-3}	10^{-4}	100
	C6	1	8	1	10^{-3}	10^{-4}	100
Bent	C1	—	16	1	10^{-3}	10^{-4}	50
	C2	—	32	1	10^{-3}	10^{-4}	50
	C3	1	16	1	10^{-3}	10^{-4}	100
	C4	2	16	1	10^{-3}	10^{-4}	100
	C5	1	8	1	10^{-3}	10^{-4}	100
	C6	1	16	1	10^{-3}	10^{-4}	100
Full	C1	—	32	1	10^{-3}	10^{-4}	50
	C7	—	32	2	10^{-4}	10^{-5}	50

9.2 Model Training Figures

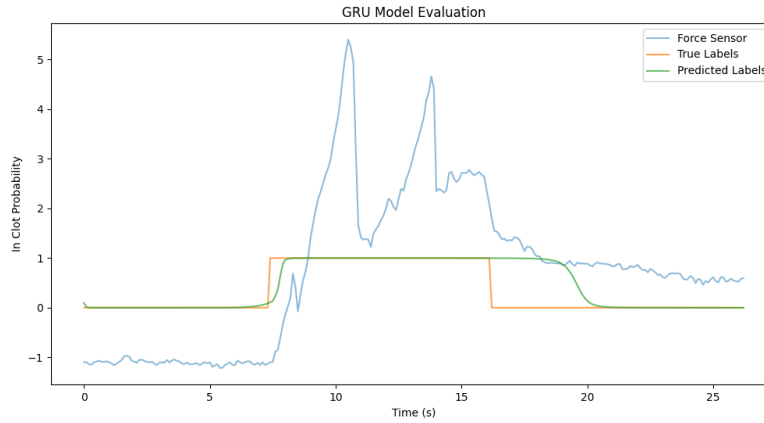
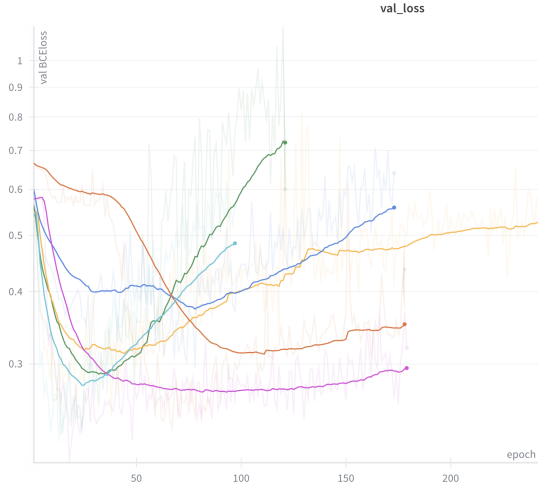
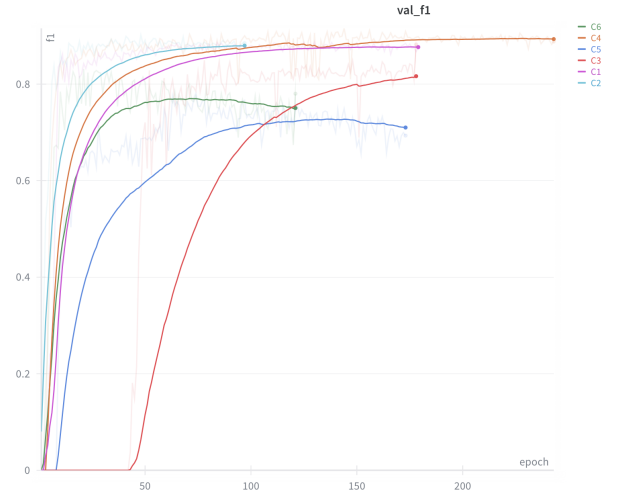


Figure 13: Example of a prediction from the C1 model train on the bent dataset on one of the sample of the base dataset. In blue, the standardized force sensor, input data of the model, in green the predicted label and in orange the true label.



(a) Evaluation loss over epochs.



(b) Evaluation F1 score over epochs.

Figure 14: Performance evolution during the bent dataset models evaluation. Left: loss curve. Right: F1 score curve.

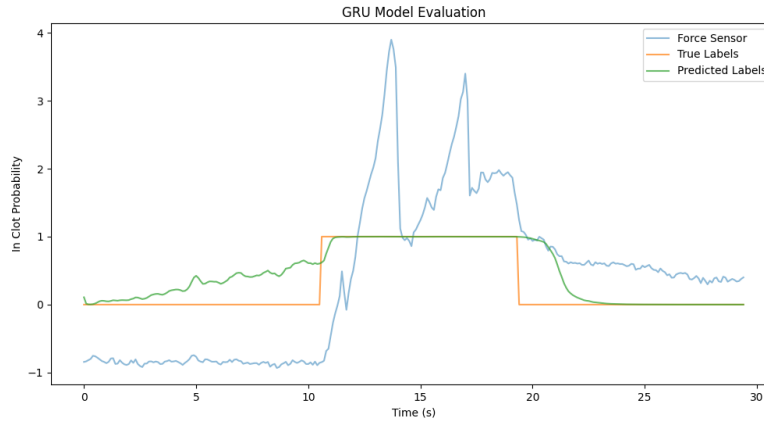


Figure 15: Example of a prediction from the C1 model trained on twist data on one of the sample of the base dataset. In blue, the standardized force sensor, input data of the model, in green the predicted label and in orange the true label.

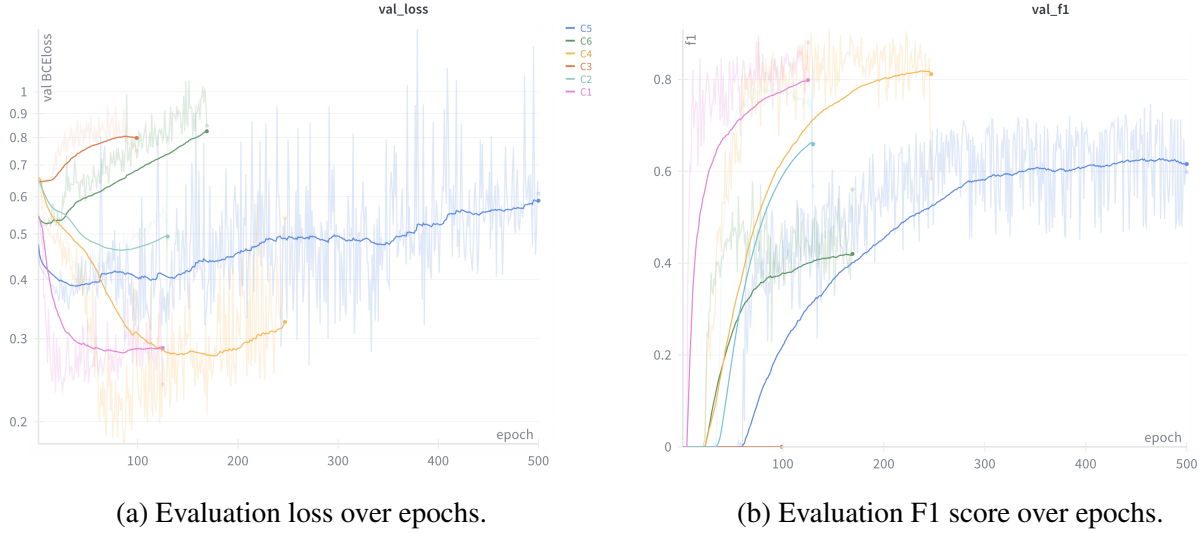


Figure 16: Performance evolution during the twist dataset models evaluation. Left: loss curve. Right: F1 score curve.

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